A

Student Table:

Student_ID	Stu_Name	Stu_Subject _ID	Stu_Marks	Stu_Age
101	Akhil	BCA101	85	20
102	Balram	BCA104	78	19
103	Bheem	BCA102	80	22
104	Chetan	BCA103	95	20
105	Diksha	BCA104	99	20
106	Raman	BCA105	88	19
107	Sheetal	BCA103	98	22

Subject Table:

Subject_ID	Subject_Name
BCA101	C
BCA102	C++
BCA103	Principle of Management
BCA104	Core Java
BCA105	Math
BCA106	Android

Query 1: Write a query to create the table in Structured Query Language.

Query 2: Write a query to insert the data into the table.

Query 3: Write a query to view the specific record of the table by using the WHERE clause.

Query 4: Write a query in SQL to find the minimum and maximum number from the integer column:

Query 5: Write a query to access the first record from the SQL table?

Query 6: Write a query to access the last record from the table?

Query 7: Write a query to access the first Nth rows from the table?

Query 8: Write a query to access the last Nth rows from the SQL table?

Query 9: Write a query to show the record of those students whose name begins with the 'm' character

Query 10: Write a query in structured query language to view all student details from the Student table order by Stu_Name Descending.

##B

SQL Query Table (Employee):

EmployeeID	FirstName	LastName	Department	Salary
1	John	Smith	HR	50000
2	Jane	Doe	IT	60000
3	Bob	Johnson	Finance	55000
4	Mary	Brown	IT	62000
5	Alice	White	Sales	48000

SQL Questions :

1. ***Question**: Retrieve the first and last names of all employees.

2. ***Question:** List the employees who work in the IT department.

3. ***Question**: Calculate the average salary of all employees.

4. ***Question**: Find the highest salary among all employees.

5. *Question:* Count the number of employees in each department.

6. *Question:* Retrieve the employees with salaries greater than $55,000 and in the HR or Finance department.

7. *Question:* List the employees with names containing "a" in their first name.

8. *Question:* Calculate the total salary expenditure for the company.

9. *Question:* Find the employee with the lowest salary.

10. *Question:* Update the salary of Bob Johnson to $56000

##C

Sample Table – Worker

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT
001	Monika	Arora	100000	2014-02-20 09:00:00	HR
002	Niharika	Verma	80000	2014-06-11 09:00:00	Admin
003	Vishal	Singhal	300000	2014-02-20 09:00:00	HR
004	Amitabh	Singh	500000	2014-02-20 09:00:00	Admin
005	Vivek	Bhati	500000	2014-06-11 09:00:00	Admin
006	Vipul	Diwan	200000	2014-06-11 09:00:00	Account
007	Satish	Kumar	75000	2014-01-20 09:00:00	Account
008	Geetika	Chauhan	90000	2014-04-11 09:00:00	Admin

Sample Table – Bonus

WORKER_REF_ID	BONUS_DATE	BONUS_AMOUNT
1	2016-02-20 00:00:00	5000
2	2016-06-11 00:00:00	3000
3	2016-02-20 00:00:00	4000
1	2016-02-20 00:00:00	4500
2	2016-06-11 00:00:00	3500

Sample Table – Title

WORKER_REF_ID	WORKER_TITLE	AFFECTED_FROM
1	Manager	2016-02-20 00:00:00
2	Executive	2016-06-11 00:00:00
8	Executive	2016-06-11 00:00:00
5	Manager	2016-06-11 00:00:00
4	Asst. Manager	2016-06-11 00:00:00
7	Executive	2016-06-11 00:00:00
6	Lead	2016-06-11 00:00:00
3	Lead	2016-06-11 00:00:00

Q-1. Write an SQL query to fetch “FIRST_NAME” from the Worker table using the alias name <WORKER_NAME>.

Q-2. Write an SQL query to fetch “FIRST_NAME” from the Worker table in upper case.

Q-3. Write an SQL query to fetch unique values of DEPARTMENT from the Worker table.

Q-4. Write an SQL query to print all Worker details from the Worker table order by FIRST_NAME Ascending.

Q-5. Write an SQL query to print all Worker details from the Worker table order by FIRST_NAME Ascending and DEPARTMENT Descending.

Q-6. Write an SQL query to print details for Workers with the first names “Vipul” and “Satish” from the Worker table.

Q-7. Write an SQL query to print details of workers excluding first names, “Vipul” and “Satish” from the Worker table.

Q-8. Write an SQL query to print details of the Workers whose FIRST_NAME contains ‘a’.

Q-9. Write an SQL query to print details of the Workers whose SALARY lies between 100000 and 500000

Q-10. Write an SQL query to fetch the no. of workers for each department in descending order.

##D

1. How do you select all columns from a table named "employees"?
   Answer: `SELECT * FROM employees;`
2. What is the SQL command to retrieve unique values from a column named "city" in a table named "customers"?

   Answer: `SELECT DISTINCT city FROM customers;`
3. How do you count the number of records in a table named "orders"?
   Answer: `SELECT COUNT(*) FROM orders;`
4. What is the SQL command to retrieve all orders placed by a specific customer with the ID 123?
   Answer: `SELECT * FROM orders WHERE customer_id = 123;`
5. How do you retrieve the first 10 records from a table named "products"?
   Answer: `SELECT * FROM products LIMIT 10;`
6. What is the SQL command to update the "price" column of a product with ID 456 to $20.99?
   Answer: `UPDATE products SET price = 20.99 WHERE product_id =`

456;

7. How can you delete all records from a table named "employees" where the "department" is "HR"?

   Answer: DELETE FROM employees WHERE department = 'HR';

8. What is the SQL command to retrieve the average salary of all employees in the "salary" column of a table named "employees"?

Answer: SELECT AVG(salary) FROM employees;

9. How do you retrieve the names of customers whose names start with the letter "A"?

   Answer: SELECT name FROM customers WHERE name LIKE 'A%';

10. What is the SQL command to join two tables "orders" and "customers" on the "customer_id" column?

Answer: SELECT * FROM orders INNER JOIN customers ON orders.customer_id = customers.customer_id;

##E

Employee_num	Employee_name	Department	Salary
1	Amit	CSE	680000
2	Rishi	CSE	550000
3	Prianka	CSE	430000

1. Write a query to create the table in Structured Query Language.
2. Write a query to insert data into the table.
3. Write a query to view the specific record of the table by using the WHERE clause?

4. Write a query that access the first the second record from the SQL table?
5. Write a query to find second highest Salary of Employee?
6. Write a query how to find monthly salary of Employee if annual salary is given?
7. Write an SQL query to find the maximum, minimum, and average salary of the employees.
8. Write an SQL query to find the employee id whose salary lies in the range of 600000 and 690000.
9. . Write an SQL query to fetch common records between two tables.
10. Write an SQL query to fetch records that are present in one table but not in another table.

##F

StudentID	FirstName	LastName	Age	GPA	Major
10001	John	Doe	20	3.75	Computer Science
10002	Jane	Smith	21	3.90	Mathematics
10003	Alice	Johnson	22	3.60	Physics
10004	Bob	Wilson	19	3.45	Chemistry
10005	Emily	Brown	20	3.85	Biology
10006	Sarah	Lee	21	3.70	Computer Science
10007	David	Anderson	22	3.55	Mathematics

Student Table Questions:

1. How do you retrieve all records from the 'Student' table?
2. What SQL statement would you use to filter students who are older than 20?
3. How do you retrieve students whose first name starts with 'A'?
4. What SQL statement would you use to update the GPA of a student with 'StudentID' 10002?
5. How do you delete a student with 'StudentID' 10005 from the 'Student' table?
6. What is the purpose of an index, and how can you create an index on the 'Major' column?
7. Write an SQL query to find the student id whose age lies in the range of 20 and 21.
8. Write an SQL query to fetch the last five records from a table
9. Retrieve the student(s) with the highest GPA
10. How would you insert a new student record into the 'Student' table?

##G

EmployeePosition table:

EmpID	EmpPosition	DateOfJoining	Salary
1	Executive	01/04/2020	75000
2	Manager	03/04/2020	500000
3	Manager	02/04/2020	150000
2	Officer	02/04/2020	90000
1	Manager	03/04/2020	300000

EmpID	EmpFname	EmpLname	Department	Project	Address	DOB	Gender
1	Rohit	Gupta	Admin	P1	Delhi	02/12/1979	Male
2	Rahul	Mahajan	Admin	P2	Mumbai	10/10/1986	Male
3	Sonia	Banerjee	HR	P3	Pune	05/06/1983	Female
4	Ankita	Kapoor	HR	P4	Chennai	28/11/1983	Female
5	Swati	Garg	HR	P5	Delhi	06/04/1991	Female

1.Write a query to get the EmpFname from the EmployeeInfo table in the upper case using the alias name as EmpName.

2.Write a query to get the number of employees working in the department ‘HR’.

3.What query will you write to fetch the current date?

4.Write a query to fetch only the place name(string before brackets) from the Address column of the EmployeeInfo table.

5.Write a query to create a new table whose data and structure are copied from another table.

6.Write a query to display the names of employees that begin with ‘S’.

7.Create a query to obtain display employees having salaries equal to or greater than 150000.

8.Write a query to fetch the year using a date.

9.Write a query to find duplicate records from a table.

10.Create a query to fetch the list of employees of the same department.

##H

Empid	EmpName	Department	ContactNo	EmailId	EmpHeadId
101	Isha	E-101	1234567890	isha@gmail.com	105
102	Priya	E-104	1234567890	priya@yahoo.com	103
103	Neha	E-101	1234567890	neha@gmail.com	101

104	Rahul	E-102	1234567890	rahul@yahoo.com	105
105	Abhishek	E-101	1234567890	abhishek@gmail.com	102

Queries:-

1. Select the detail of the employee whose name start with P.
2. Select the detail of employee whose emailId is in gmail.
3. Select the details of the employee who work either for department E-104 or E-102.
4. What is the department name for DeptID E-102?
5. List name of all employees whose name ends with a.

EmployeeInfo Table:

EmpID	EmpFname	EmpLname	Department	Project	Address	DOB	Gender
1	Sanjay	Mehra	HR	P1	Hyderabad(HYD)	01/12/1976	M
2	Ananya	Mishra	Admin	P2	Delhi(DEL)	02/05/1968	F
3	Rohan	Diwan	Account	P3	Mumbai(BOM)	01/01/1980	M
4	Sonia	Kulkarni	HR	P1	Hyderabad(HYD)	02/05/1992	F
5	Ankit	Kapoor	Admin	P2	Delhi(DEL)	03/07/1994	M

EmployeePosition Table:

EmpID	EmpPosition	DateOfJoining	Salary
1	Manager	01/05/2022	500000
2	Executive	02/05/2022	75000
3	Manager	01/05/2022	90000
2	Lead	02/05/2022	85000
1	Executive	01/05/2022	300000

6. Write a query to find the Nth highest salary from the table without using TOP/limit keyword.

7.Write a query to retrieve duplicate records from a table.

8.Write a query to display the first and the last record from the EmployeeInfo table.

9. Write a query to retrieve two minimum and maximum salaries from the EmployeePosition table.

10.Write a query to calculate the even and odd records from a table.

##I

```
CREATE DATABASE student_inquery;

SHOW DATABASES;

USE student_inquery;


CREATE TABLE student (
        Student_id INT(8) NOT NULL PRIMARY KEY,
        First_name CHAR(25),
        Last_name CHAR(25),
        Money INT(15),
    Money_source char(15),
    Monthley_expenses int(8),
        Department CHAR(25)
);


INSERT INTO  student
```

(Student_id, First_name, Last_name,Money, Money_source, Monthley_expenses,Department) VALUES

(2002040, 'Shabuj', 'Mia',8000, 'Tution',7500, 'CSE'),

(2004067, 'Sorna', 'Boikontho', 5000, 'Home',4500, 'DVM'),

(2002022, 'Ronok', 'Hasan', 300000, 'Freelancing',11000 ,'CSE'),

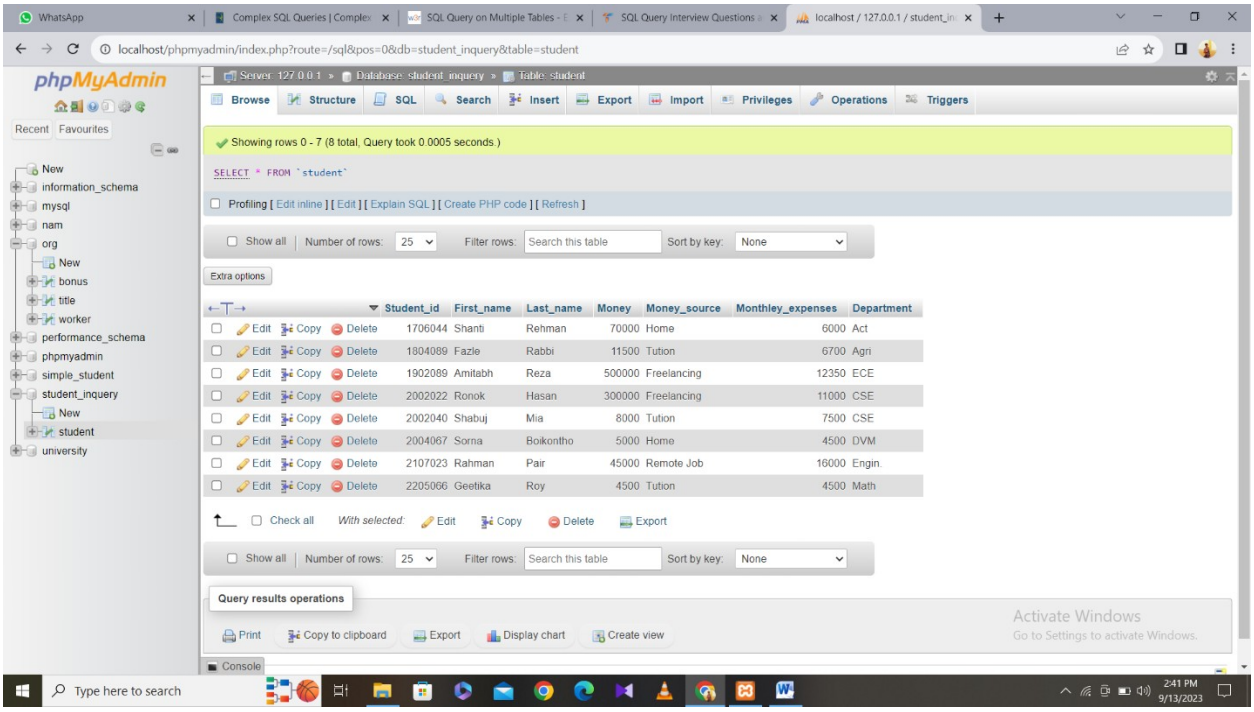
(1902089, 'Amitabh', 'Reza', 500000, 'Freelancing',12350, 'ECE'),

(1804089, 'Fazle', 'Rabbi', 11500, 'Tution',6700, 'Agri'),

(1706044, 'Shanti', 'Rehman',70000, 'Home',6000, 'Act'),

(2107023, 'Rahman', 'Pair', 45000, 'Remote Job',16000, 'Engin.'),

(2205066, 'Geetika', 'Roy', 4500, 'Tution',4500, 'Math');



Q(1): What is SQL Query ? Why it is needed ? write it's application.

Q(2): Write a SQL query to print First_name and Last_name from table into a single column Full_name.A space char should separate them.

Q(3): Write a SQL query to find those student money is greater than 10000 and expenses is greater than 10000.

Q(4): Write a SQL query to find those student expenses is smaller than 70000.

Q(5): Write a SQL query to find unique money source.

Q(6): Write a SQL query to find highest money in freelancing.

Q(7): Write a SQL query to find highest money in different job. Like (Home 7000, Freelancing 50000)

Q(8): Write a SQL query to find details for First_name “Ronok” Last_name “Hasan” from the table.

Q(9): Write a SQL query to find even Student_id.

Q(10): write a SQL query to find those student whose First_name start with “A” or “S”.

##J

WORKER_ID	FIRST_NAME	LAST_NAME	SALARY	JOINING_DATE	DEPARTMENT

001	Monika	Arora	100000	2014-02-20 09:00:00	HR
002	Niharika	Verma	80000	2014-06-11 09:00:00	Admin
003	Vishal	Singhal	300000	2014-02-20 09:00:00	HR
004	Amitabh	Singh	500000	2014-02-20 09:00:00	Admin
005	Vivek	Bhati	500000	2014-06-11 09:00:00	Admin
006	Vipul	Diwan	200000	2014-06-11 09:00:00	Account
007	Satish	Kumar	75000	2014-01-20 09:00:00	Account
008	Geetika	Chauhan	90000	2014-04-11 09:00:00	Admin

Sample Table – Worker

Sample Table – Bonus

WORKER_REF_ID	BONUS_DATE	BONUS_AMOUNT
1	2016-02-20 00:00:00	5000
2	2016-06-11 00:00:00	3000
3	2016-02-20	4000

WORKER_REF_ID	BONUS_DATE	BONUS_AMOUNT
	00:00:00	
1	2016-02-20 00:00:00	4500
2	2016-06-11 00:00:00	3500

Sample Table – Title

WORKER_REF_ID	WORKER_TITLE	AFFECTED_FROM
1	Manager	2016-02-20 00:00:00
2	Executive	2016-06-11 00:00:00
8	Executive	2016-06-11 00:00:00
5	Manager	2016-06-11 00:00:00
4	Asst. Manager	2016-06-11 00:00:00
7	Executive	2016-06-11 00:00:00
6	Lead	2016-06-11 00:00:00
3	Lead	2016-06-11 00:00:00

1.Write an SQL query to print all Worker details from the Worker table order by FIRST_NAME Ascending and DEPARTMENT Descending.

2.Write an SQL query to print details of Workers with DEPARTMENT name as "Admin".

3. Write an SQL query to print details of workers excluding first names, "Vipul" and "Satish" from the Worker table.

4.Write an SQL query to print details of the Workers whose SALARY lies between 100000 and 500000.

5. Write an SQL query to fetch the no. of workers for each department in descending order.

6.Write an SQL query to fetch duplicate records having matching data in some fields of a table.

7. Write an SQL query to show only even rows from a table.

8. Write an SQL query to fetch intersecting records of two tables.

9.Write an SQL query to show records from one table that another table does not have.

10.Write an SQL query to show the top n (say 10) records of a table.